MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **GENERAL ARRANGEMENT DRAWING**
FUEL OIL STORAGE FACILITY
LAYOUT & DETAILS OF CABLE PULL PITS

Document Number: **MD-216-49FE-EG-CV-GAD-1639**
Revision: **D01**
Revision Date: **06.09.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AG50	MTL01	49056	OU100722079	3
L&T Document Number:	O19045	080	C	G	24550	001	3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	25.05.2021	ISSUED FOR APPROVAL	SYA	SJM	AG	
A02	17.07.2021	REVISED AS PER EMPLOYER COMMENTS DATED 22.06.2021	SYA	SJM	AG	
C01	13.09.2021	ISSUED FOR CONSTRUCTION	SYA	SJM	AG	
D01	06.09.2023	AS - BUILT	SYA	SJM	AG	

DRAWING NUMBER: MD - 216 - 49FE - EG - CV - GAD - 1639 | REV: D01

FOR BURIED CABLE REFER DRAWING NO MD-216-3000-EG-EL-LAD-1021

E = 263853.879
N = 2505970.721

PULL PIT CP-001B

REFER DRAWING NO MD-216-49PR-EG-CV-GAD-2031

GRID NORTH (PROJECT NORTH) 0°-14'-32" EAST OF TRUE NORTH

52°-58'-00" MAGNETIC NORTH

328

CB-R5

E= 263947.593
N=2505971.703

E= 263937.310
N=2505954.545

E= 263970.314
N=2505985901

CB-R1

E= 263899.530
N=2505914.940

PULL PIT CP-011

E= 263896.444
N=2505906.615

E = 263879.691
N = 2505896.147

CB-R2

PULL PIT CP-001A

E= 263944.358
N=2505943.240

39WQ

REFER DRAWING NO MD-216-39WQ-EG-CV-GAD-1001

E = 263973.884
N = 2505927.833

E = 263990.845
N = 2505938.431

CB-R3

PULL PIT CP-016

FOR BURIED CABLE REFER DRAWING NO MD-216-3000-EG-EL-LAD-1021

REFER DRAWING NO MD-216-49FE-EG-CV-GAD-1631

PULL PIT CP-012

E = 263946.662
N = 2505893.860

BURIED CABLE (TYP.)

4268

5536

LPG VAPORIZER ROOM

FOR BURIED CABLE REFER DRAWING NO MD-216-3000-EG-EL-LAD-1021

49FE

LPG BULLET

PULL PIT CP-013

E = 263947.680
N = 2505869.615

LPG COMPRESSOR ROOM

E = 263945.392
N = 2505848.436

CB-R4

PULL PIT CP-015

E = 263949.891
N = 2505841.237

REFER DRAWING NO MD-216-49FE-EG-CV-GAD-1636

LPG UNLOADING OPERATOR ROOM

LPG STORAGE AREA PLAN

E= 264011.965
N=2505868.902

CB-R6

E= 264025.966
N=2505877.651

PULL PIT CP-012

BURIED CABLE ROUTE

TO LPG STORAGE AREA

LPG VAPORIZER ROOM

BURIED CABLE ROUTE

PULL PIT CP-013

LPG COMPRESSOR ROOM

CONCRETE ENCASED CONDUIT BANK

PULL PIT CP-015

LPG UNLOADING OPERATOR ROOM

PULL PIT CP-002A

E = 263859.594
N = 2505831.273

MINE WATER OPERATOR ROOM

MINE WATER OPERATOR ROOM

R.C. SECTION 1-1
SCALE 1:30

GRID NORTH (PROJECT NORTH) 0°-14'-32" EAST OF TRUE NORTH

MAGNETIC NORTH

TANK

2 NOS PETACH DISPENSING PLATFORM WITH STRAINER TYP.

CAR WASHING STATION

MMGP-5

MINE WATER OPERATOR BRIDGE

LPG UNLOADING STATION

KEY PLAN

VIEW 05-05
SCALE 1:75

G.A. SECTION A4-A4
SCALE 1:30

G.A. SECTION A5-A5
SCALE 1:30

G.A. SECTION T1-T1
SCALE 1:30

G.A. SECTION Q2-Q2
SCALE 1:30

R.C. PLAN FOR PULLPIT
SCALE 1:30

G.A. PLAN FOR PULLPIT MKD. CP-012
SCALE 1:30

G.A. PLAN FOR PULLPIT MKD. CP-016
SCALE 1:30

G.A. SECTION Q1-Q1
SCALE 1:30

G.A. SECTION Q4-Q4
SCALE 1:30

G.A. SECTION Q5-Q5
SCALE 1:30

G.A. SECTION Q3-Q3
SCALE 1:30

G.A. PLAN FOR PULLPIT MKD. CP-013
SCALE 1:30

G.A. SECTION T4-T4
SCALE 1:30

G.A. PLAN FOR PULLPIT MKD. CP-015
SCALE 1:30

G.A. PLAN FOR PULLPIT MKD. CP-001A
SCALE 1:30

DET. OF CB-R1, CB-R5 & CB-R6
SCALE 1:25

G.A. PLAN FOR PULLPIT MKD. CP-011
SCALE 1:30

G.A. PLAN FOR PULLPIT MKD. CP-001B
SCALE 1:30

SECTION CB-R2
SCALE 1:25

SECTION CB-R3
SCALE 1:25

SECTION CB-R4
SCALE 1:25

LEGEND:

TYP.	TYPICAL	T	TOP
U.N.O	UNLESS NOTED OTHERWISE	B	BOTTOM
G.A.	GENERAL ARRANGEMENT		
R.C.	REINFORCED CONCRETE	MKD.	MARKED
F.G.L	FINISHED GROUND LEVEL	CHQD.	CHEQUERED
P.C.C	PLAIN CEMENT CONCRETE	THK.	THICK
R.C.C	REINFORCED CEMENT CONCRETE	PL.	PLATE
DIA	DIAMETER	C/C	CENTRE TO CENTRE
ø	DIAMETER		

NOTES:
1. ALL DIMENSIONS ARE IN MILLIMETRES
2. CO-ORDINATES AND LEVELS ARE IN METRES
3. CO-ORDINATES ARE ON WORLD GRID (FLAT PLANT/WORLD)
4. WORLD MAPPING GRID BASED ON UTM (WGS 84) 038N
5. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF PITS IS CONSIDERED AS 10T/ SQ. M.
6. GRADE OF REINFORCED CEMENT CONCRETE (R.C.C) SHALL BE M35.
7. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)
8. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE 25mm(MINIMUM) U.N.O.
9. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-49FE-EG-CV-GAD-1639 (SHEET 2 OF 3) & (SHEET 3 OF 3).
10. FOR CONCRETE GENERAL NOTES, REFER DRAWING NUMBER MD-216-1000-EG-CV-DED-1005.

SHEET 1 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	MD-216-3000-EG-EL-LAD-1021	A02	21/07/2021	REVISED AS PER EMPLOYER COMMENTS DATED 22/06/2021	SYA	SJM	AVG	AG	
	MD-216-49FE-EG-EL-LAD-1005	A01	25/05/2021	ISSUED FOR APPROVAL	SYA	SJM	AVG	AG	
		D01	06/09/2023	AS-BUILT	SYA	SJM	AVG	AG	
		C01	13/09/2021	ISSUED FOR CONSTRUCTION	SYA	SJM	AVG	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

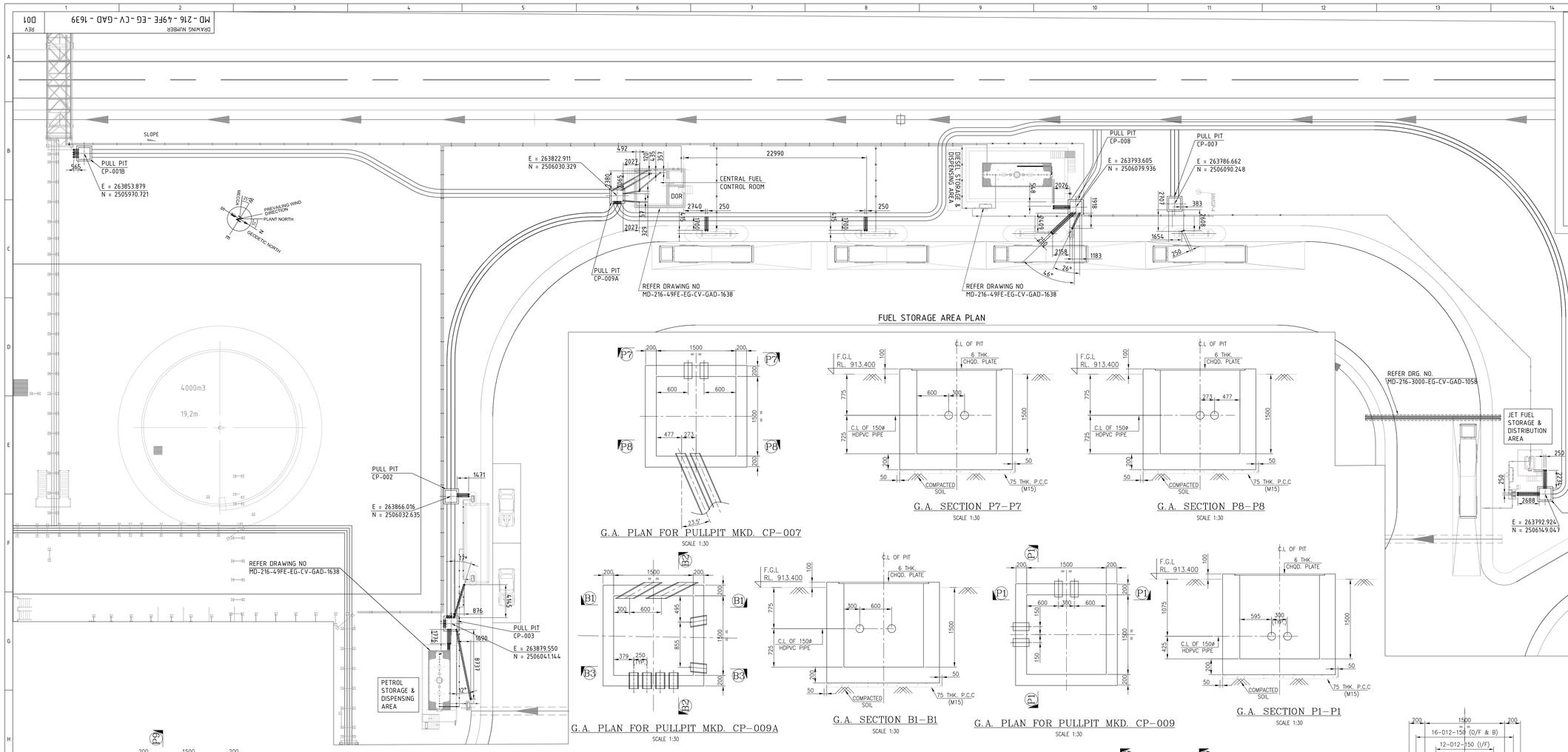
LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_080_C_G_24550_001	3
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 __ GAU01 __ AG50_ MTL01 __49056	OU110722079 / 3

DRAWN	SYA	10/05/2021
CHECKED	SJM	11/05/2021
ENGINEER	AVG	10/05/2021
APPROVED	AG	25/05/2021
EMPLOYER		
SCALE	1:200	A0

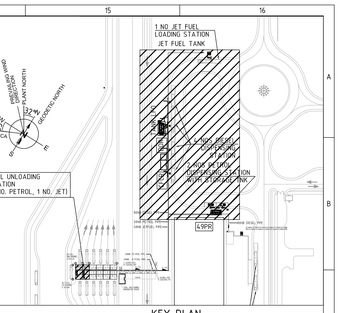
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
FUEL OIL STORAGE FACILITY
LAYOUT & DETAILS OF CABLE PULL PITS

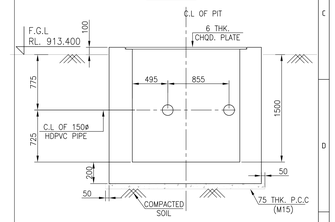
DRAWING NUMBER: MD - 216 - 49FE - EG - CV - GAD - 1639 | REV: D01

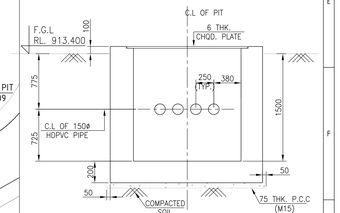
DRAWING NUMBER: MD-216-49FE-EG-CV-GAD-1639 | REV: D01

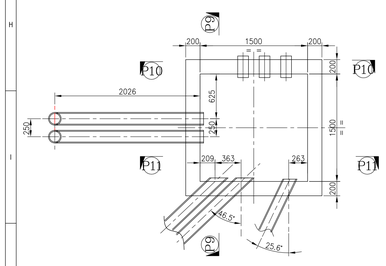


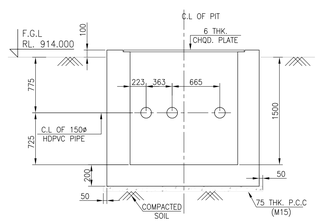
FUEL STORAGE AREA PLAN

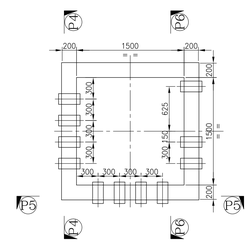


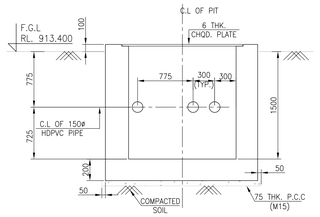


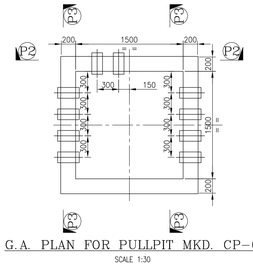


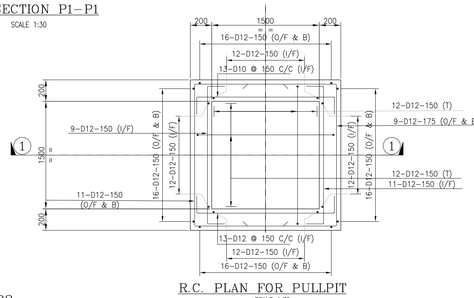


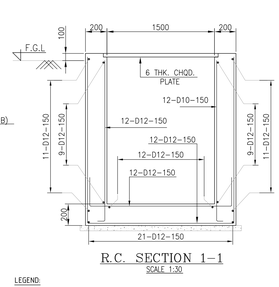


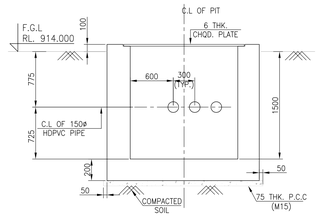


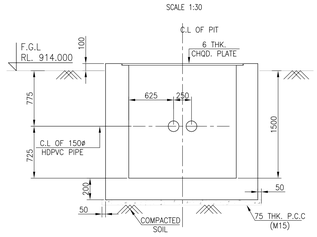


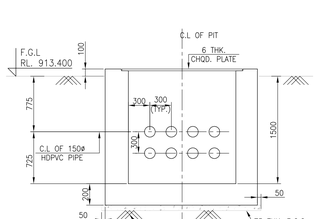


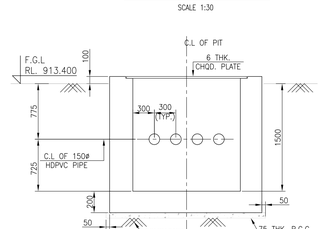


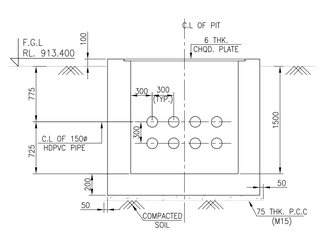


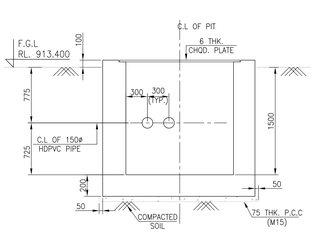












LEGEND:

TYP.	TYPICAL
U.N.O	UNLESS NOTED OTHERWISE
G.A.	GENERAL ARRANGEMENT
R.C.	REINFORCED CONCRETE
F.G.L	FINISHED GROUND LEVEL
P.C.C	PLAIN CEMENT CONCRETE
R.C.C	REINFORCED CEMENT CONCRETE
DIA	DIAMETER
ø	DIAMETER
T	TOP
B	BOTTOM
MKD.	MARKED
CHQD.	CHEQUERED
THK.	THICK
PL.	PLATE
C/C	CENTRE TO CENTRE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES
2. CO-ORDINATES AND LEVELS ARE IN METRES
3. CO-ORDINATES ARE ON WORLD GRID (FLAT PLANT/WORLD)
4. WORLD MAPPING GRID BASED ON UTM (WGS 84) 038N
5. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF PITS IS CONSIDERED AS 10T/ SQ. M.
6. GRADE OF REINFORCED CEMENT CONCRETE (R.C.C) SHALL BE M35.
7. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)
8. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE 25mm(MINIMUM) U.N.O.
9. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-49FE-EG-CV-GAD-1639 (SHEET 1 OF 3) & (SHEET 3 OF 3).
10. FOR CONCRETE GENERAL NOTES, REFER DRAWING NUMBER MD-216-1000-EG-CV-DED-1005.

SHEET 2 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-EL-LAD-1021	PROCESS PLANT UNDER GROUND CABLE ROUTE DETAILS	A02	21/07/2021	REVISED AS PER EMPLOYER COMMENTS DATED 22/06/2021	SYA	SJM	AVG	AG	
MD-216-49FE-EG-EL-LAD-1002	FUEL OIL STORAGE CABLE ROUTE LAYOUT	A01	25/05/2021	ISSUED FOR APPROVAL	SYA	SJM	AVG	AG	
		D01	06/09/2023	AS-BUILT					
		C01	13/09/2021	ISSUED FOR CONSTRUCTION	SYA	SJM	AVG	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

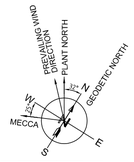
LARSEN & TOUBRO
This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use of/we in full or in part by such person or body without our prior permission in writing.

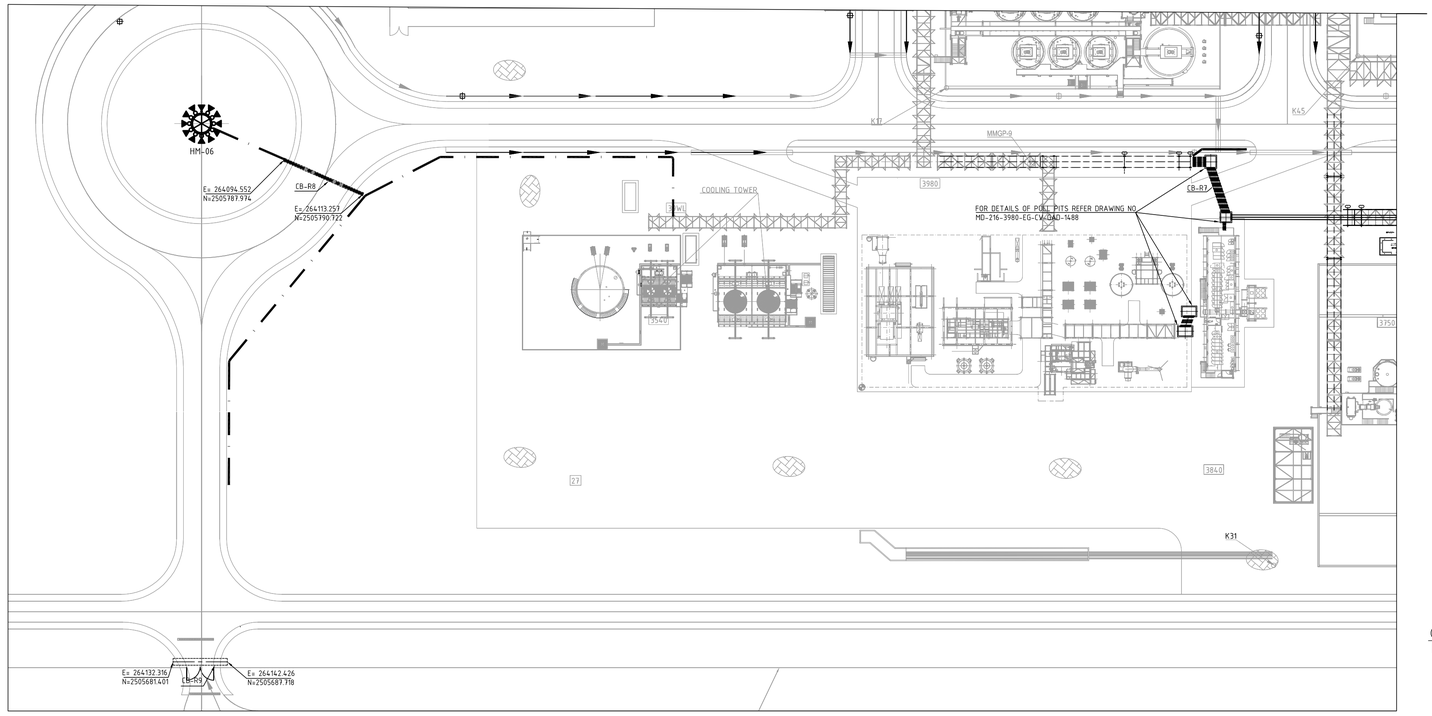
Outotec
The content of these documents are the sole property of Outotec Oy. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

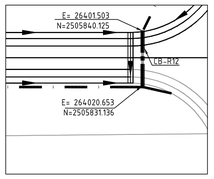
MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

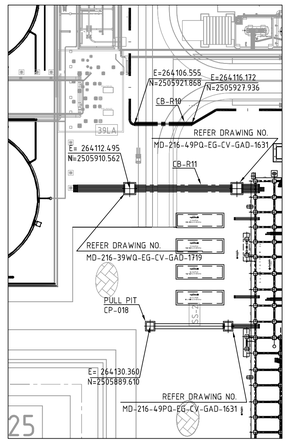
LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_080_C_G_24550_001			3
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__AG50__MTL01__49056		OU100722079	3
DRAWN	SYA	10/05/2021	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SJM	11/05/2021	GENERAL ARRANGEMENT DRAWING
ENGINEER	AVG	10/05/2021	FUEL OIL STORAGE FACILITY
APPROVED	AG	25/05/2021	LAYOUT & DETAILS OF CABLE PULL PITS
EMPLOYER			
SCALE 1:200		A0	DRAWING NUMBER: MD-216-49FE-EG-CV-GAD-1639 | REV: D01

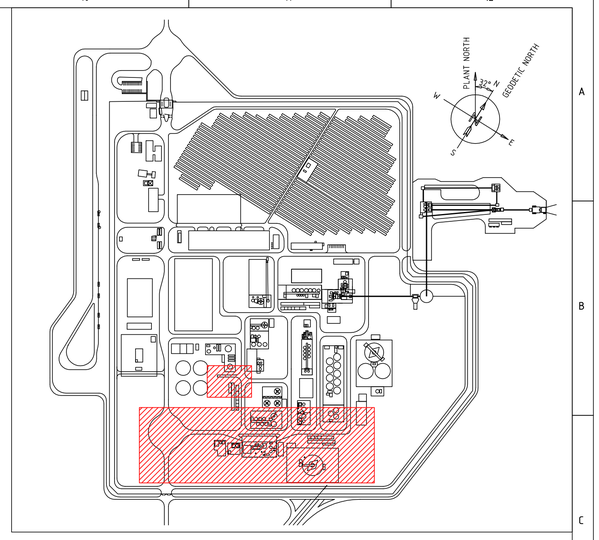
DRAWING NUMBER
MD - 216 - 49FE - EG - CV - GAD - 1639
REV D01



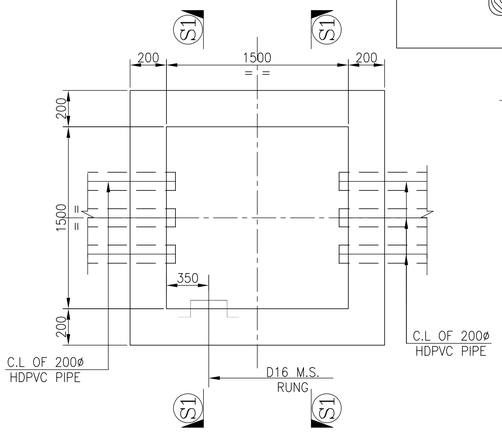




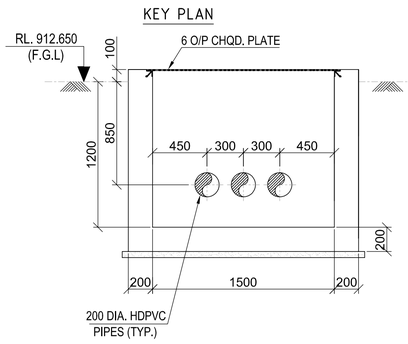




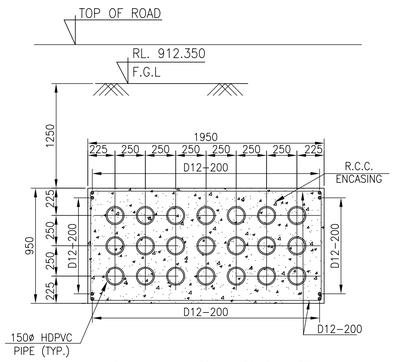
KEY PLAN



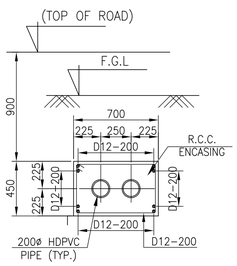
G.A. PLAN FOR PULLPIT MKD. CP-018
SCALE 1:30



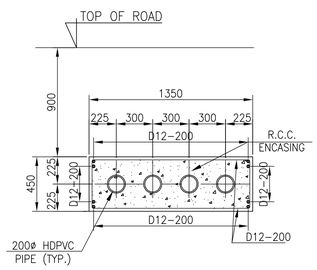
G.A. SECTION S1-S1
SCALE 1:30



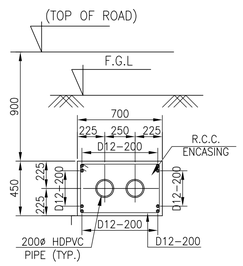
DETAIL OF CB-R7
SCALE 1:25



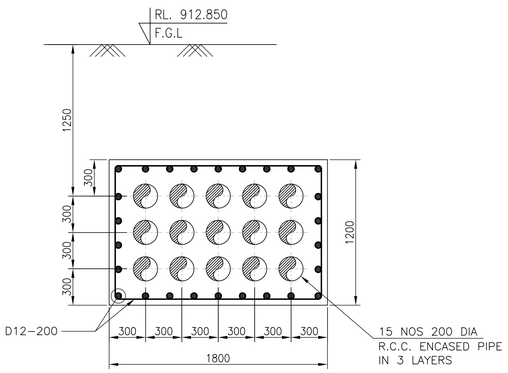
DET. OF CB-R8
SCALE 1:25



DETAIL OF CB-R9
SCALE 1:25



DET. OF CB-R10 & CB-R12
SCALE 1:25



DETAIL OF CB-R11
SCALE 1:25

LEGEND:

TYP.	TYPICAL
U.N.O	UNLESS NOTED OTHERWISE
G.A.	GENERAL ARRANGEMENT
R.C.	REINFORCED CONCRETE
F.G.L	FINISHED GROUND LEVEL
P.C.C	PLAIN CEMENT CONCRETE
R.C.C	REINFORCED CEMENT CONCRETE
DIA	DIAMETER
ø	DIAMETER
T	TOP
B	BOTTOM
MKD.	MARKED
CHQD.	CHEQUERED
THK.	THICK
PL.	PLATE
C/C	CENTRE TO CENTRE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES
2. CO-ORDINATES AND LEVELS ARE IN METRES
3. CO-ORDINATES ARE ON WORLD GRID (FLAT PLANT/WORLD)
4. WORLD MAPPING GRID BASED ON UTM (WGS 84) 038N
5. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF PITS IS CONSIDERED AS 10T/ SQ. M.
6. GRADE OF REINFORCED CEMENT CONCRETE (R.C.C) SHALL BE M35.
7. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)
8. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE 25mm(MINIMUM) U.N.O.
9. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-49FE-EG-CV-GAD-1639 (SHEET 1 OF 3) & (SHEET 2 OF 3).
10. FOR CONCRETE GENERAL NOTES, REFER DRAWING NUMBER MD-216-1000-EG-CV-DED-1005.

SHEET 3 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS			REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	06/09/2023	AS-BUILT					
		C01	13/09/2021	ISSUED FOR CONSTRUCTION	SYA	SJM	AVG	AG	
		A02	21/07/2021	REVISED AS PER EMPLOYER COMMENTS DATED 22/06/2021	SYA	SJM	AVG	AG	
MD-216-3000-EG-EL-LAD-1021	PROCESS PLANT UNDER GROUND CABLE ROUTE DETAILS	A01	25/05/2021	ISSUED FOR APPROVAL	SYA	SJM	AVG	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO
This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec
The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWN	SYA	10/05/2021
CHECKED	SJM	11/05/2021
ENGINEER	AVG	10/05/2021
APPROVED	AG	25/05/2021
EMPLOYER		
SCALE	1:150, 50, 40, 30	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_C_G_24550_001 — REVISION: 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49056 — DOCUMENT ID: OU100723932 — REVISION: 3

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
FUEL OIL STORAGE FACILITY
LAYOUT & DETAILS OF CABLE PULL PITS

DRAWING NUMBER: MD - 216 - 49FE - EG - CV - GAD - 1639 — REV: D01